MANUEL PAEZ

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Education

Columbia University New York, NY
Master of Science, Computer Science, GPA: 3.83/4.0 Jan. 2025 - Expected Dec. 2026
Bachelor of Arts, Mathematics Sept. 2023 - May 2026
Bachelor of Arts, Computer Science Sept. 2019 - May 2023
Relevant Coursework: Advanced Probability, Advanced Deep Learning, Algorithms for Massive Data, Machine Learning Theory, Log-Concave Probability, Statistical Inference, Theoretical Foundations of LLMs, Unsupervised Learning.

Work and Research Experience

Columbia University New York, NY
Course Assistant, Department of Computer Science Sept. 2024 - Dec. 2025

- Organized Recitations, held office hours, and graded theoretical and programming assignments for 40+ students in COMS 4995: Randomized Algorithms and COMS 6998: Algorithms for Massive Data.

Flatiron Institute, Simons Foundation New York, NY
Visiting Scholar Fellow, Neural Circuits and Algorithms Group Nov. 2024 - Aug. 2025

- Worked on self-supervised predictive representation learning for high-dimensional temporal data, using ReSU architectures and nonlinear-system methods to learn low-dimensional representations. Code primarily in Python.
- Implemented and evaluated offline/online algorithms on neuronal recordings and synthetic stochastic trajectories.

Research Intern, Neural Circuits and Algorithms Group May 2024 - Aug. 2024

- Reconstructed CaImAn's large-scale calcium-imaging analysis pipeline in PyTorch, migrating model training, inference, and pretrained-weight conversion from TensorFlow while reproducing model behavior across frameworks.
- Reintegrated an automated and scalable voltage-imaging analysis pipeline with a PyTorch backend; trained and evaluated models on 10+ TB of calcium-imaging data using Flatiron's HPC/Slurm infrastructure.

Research Intern, Neural Circuits and Algorithms Group Sept. 2022 - Aug. 2023

- Extended NeuTorch to train and evaluate a supervised IsoRSUNet for neuron-boundary segmentation, using 2.1B annotated voxels from three wasp specimens and reducing Rand-based split error by up to 52% versus baseline.
- Leveraged Flatiron's HPC cluster to automate large-scale segmentation workflows. Code primarily in Python.

Selected Research and Technical Project

Fast Geometric Spanners via Approximate Nearest Neighbor Search *Accepted to ITCS 2027*

- Developed randomized graph-spanner constructions using approximate batched nearest-neighbor search, achieving O(ck) distortion with polylog edges; proved matching $\Omega(ck)$lower bounds under black-box metric access.
- Extended the framework to subquadratic Wasserstein-q approximation using bounded-hop spanners.

Asymmetric Similarity Matching: A Normative Framework for Predictive Neural Computation.

- Developed offline and online Local Predictive Directions (LPD) algorithms via nonnegative asymmetric similarity matching (NASM), recovering predictive structure in stochastic dynamical systems with approximately 99.9% precision and recall. *(Submitted to Neural Computation.)*

Rank-One Spectral Normalization Accelerates Projected Feature Learning *ICML 2026 HiLD Workshop*

- Analyzed finite-width student-teacher optimization dynamics and spectral normalization effects on training.

Leadership

Columbia Undergraduate Quantum Computing Club Nov. 2022 - Jan. 2024

- Co-Founder and Co-Head; organized seminars and technical events, and managed weekly member communications.

Technical Skills

Python, C/C++, Java, MATLAB, R; PyTorch, CUDA, TensorFlow, Keras, Hugging Face Transformers, Slurm, Qiskit